Ines VATI

PhD Student in Mathematics and Computer Vision

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I am a PhD student specializing in Computer Vision and Machine Learning, with a focus on developing rapid and automated methods for medical image analysis. Throughout my academic journey, I have undertaken various scientific projects, including an industrial internship on multimodal medical image registration and a research internship at CSIRO on automated brain parcellation of cortical surfaces.

Education

PhD (CU)

Queensland University of Technology & CSIRO, Australia, 2025 - 2029

Doctor of Philosophy at the School of Engineering and Robotics at QUT. The title of my thesis is *Fast Geometric Machine Learning Methods for Brain Image Analysis.*

MVA Research Master (Mathematics, Vision, Learning)

ENS Paris-Saclay, France, 2023 - 2024

Master's degree in Computer Vision, Data Science and Machine Learning : Computational Statistics, Convex Optimization, Computational Optimal Transport, Kernel Methods, Point Cloud and 3D Modelisation.

Mathematical and Computer Science Degree

École des Ponts ParisTech at Champs-sur-Marne, France, 2020 - 2023

Engineering degree in Mathematics and Computer Science : Random processes, Software development techniques, Deep Learning, Statistics and Data Analysis, Image processing and Artificial Vision.

Preparatory class BCPST (Biology, Chemistry, Physics and Earth Sciences)

Baimbridge High School in Guadeloupe, France, 2017 - 2020

Undergraduate preparatory courses in biology, chemistry, physics, earth sciences and mathematics for the national competitive entrance exams to "Grandes Ecoles", the leading French engineering schools.

Work Experience

Postgraduate Intern in Machine Learning Research

CSIRO in Brisbane, Australia, April 2024 to October 2024

Conducted an in-depth review of the state-of-the-art methods in brain surface analysis and parcellation. Developed innovative approaches using geometric deep learning methods and spectral graph theory to enhance the analysis and parcellation of brain surfaces.

3D Image Registration Internship

Dassault Systèmes at Vélizy-Villacoublay, France, february 2023 to july 2023

Implemented and experimented with convolution-based neural networks, like VoxelMorph, for 3D medical image registration, aimed at reconstructing 3D anatomical structures of organs. Developed innovative models for learning anatomical templates, capturing the variability within a given patient population. Participated in an internal poster session, showcasing research findings and engaging with peers.

Deep Learning Internship

MaIAGE INRAE at Jouy-en-josas, France, september 2022 to january 2023

Implementation of Neural Networks for Phylodynamic Study of Epidemiological Models, Developed mathematical models of the dengue epidemic and simulated phylogenetic trees. Trained neural networks to infer epidemiological parameters of complex models. Achieved competitive results compared to the state-of-the-art. Presented my work during the lab's weekly seminar.

Academic Research projects

Kernel Methods Kaggle Challenge

Participated in the MVA course "Machine Learning with Kernel Methods" data challenge, focused on multi-class image classification on pre-processed images. Achieved 2nd on the leaderboard : https://www.kaggle.com/competitions/data-challenge-kernel-methods-2023-2024-extension.

Point Clouds Modelisation Project

Conducted a project as part of the MVA course "Point Cloud and 3D Modelisation" about semantic classification of 3D point clouds using multiscale spherical neighborhoods. Worked autonomously and submitted a summary report.

Medical Image Analysis Project

Conducted a study on the boundary-based loss function proposed by Kervadec et al. to tackle highly unbalanced segmentation. Completed a technical report writing and delivered a concise presentation.

MOPSI (MOdel Program SImulate) Project

Collaborated in a 4-months project within a team of three, focusing on machine learning and dimension reduction algorithms for molecular simulation. Developed an interactive website explaining key concepts, and allowing users to visualize and simulate molecular trajectories. Presented our work in a poster session.

Volunteer experience

EPVS Segmentation Challenge - MICCAI2024

Participated in the effort to segment Enlarged Perivascular Spaces (EPVS) in brain MRIs of different modalities from multiple sites and scanners.

Ocean Hackathon 6 - 1st price

48 hours to develop a prototype application for raising awareness about marine biodiversity using various sea-related data. Our team developed a prototype for an app that enables users to identity any fish species through photo recognition. Successfully pitched our product to a jury.

Coordinator of Street Outreach activities

Distribution of hot beverages and conversations with the homeless in Paris, team organization of charitable events.

President of the Drama Club, ENPC

Organized weekly theater workshops at Ecole des Ponts ParisTech, fostering creativity, public speaking, and self-confidence among students. Managed schedules, venues, and resources for club meetings, rehearsals, and two annual performances.

Private tutor

Homework assistance, Classes in Mathematics, Physics for High School to Bachelor students.

Computing

Linux	Python	R	Docker
Matlab	C++	Julia	Singularity

Interest

Sports

Running (Adidas 10k Paris), hiking, ENPC's handball club

Reading

Bibliographies, detective novels

Ex. : *Becoming* by Michelle Obama, *Brave New World* by Aldous Huxley

Creative works

Original video production for our Humanitarian project

@vatiines, Comment soutenir le projet Humanitaire DVP 023?, 21 Feb 2021

Languages

English (C1)

TOEIC in 2020

Spanish (B2)

Arabic (A2)

Soft skills

Thorough

Synthesis skills

Strong adaptability skills